

MATH120-24S2 Discrete Mathematics

Tutorial 3

References

- Topic 3 and 4: Arguments, Proof, and Mathematical Induction
- Sections 4.7, 5.2, and 5.3 of Epp (4th or 5th Ed.)

Work on the following problems in groups during the tutorial in week 4.

1. [B] Consider the following statement: *There is no greatest even integer.* Write a negation of it, and a proof of it by contradiction.

Solution:

A negation is: *There is a greatest even integer.* *Proof:* Suppose there is a greatest even integer, n . Then consider $n + 2$, which is greater than n and even. This contradicts our assumption, so in fact there is no such n .

2. [B] Prove or disprove: If m, n are positive integers and m, n are perfect squares, then $m + n$ is a perfect square.

Solution:

This result is not true in general. For instance, let $m = 4 = 2^2$ and $n = 1 = 1^2$ which are two positive integers that are perfect squares. Then,

$$m + n = 2^2 + 1^2 = 5$$

which is not a perfect square. This is one possible counterexample.

3. [M] Prove each of the following for all positive integers n using mathematical induction

$$(a) \quad 1^2 + 3^2 + 5^2 + \dots + (2n - 1)^2 = \frac{n(2n - 1)(2n + 1)}{3}$$

$$(b) \quad \sum_{i=1}^n i(2^i) = 2 + (n - 1)2^{n+1}$$

$$(c) \quad 2 + 4 + 6 + \dots + 2n = n^2 + n$$

Solution:

(a) Proof: Base Step: For $n = 1$, we require $1^2 = \frac{1(2(1) - 1)(2(1) + 1)}{3}$, which is true.

Induction Step: We wish to show that $S(k) \Rightarrow S(k + 1)$. That is, we assume

$$1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 = \frac{k(2k-1)(2k+1)}{3} \quad (\text{The induction hypothesis})$$

in order to show $S(k + 1)$, which is

$$1^2 + 3^2 + 5^2 + \dots + (2(k+1)-1)^2 = \frac{(k+1)(2(k+1)-1)(2(k+1)+1)}{3}$$

or more helpfully,

$$1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 + (2k+1)^2 = \frac{(k+1)(2k+1)(2k+3)}{3}.$$

Consider the left-hand side of the desired result:

$$\begin{aligned} & 1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 + (2k+1)^2 \\ &= \frac{k(2k-1)(2k+1)}{3} + (2k+1)^2 \quad \text{By the induction hypothesis.} \\ &= \frac{k(2k-1)(2k+1)}{3} + \frac{3(2k+1)(2k+1)}{3} \\ &= (k(2k-1) + 3(2k+1)) \frac{2k+1}{3} \quad * \\ &= (2k^2 - k + 6k + 3) \frac{2k+1}{3} \\ &= (2k^2 + 5k + 3) \frac{2k+1}{3} \\ &= (k+1)(2k+3) \frac{2k+1}{3} \\ &= \frac{(k+1)(2k+1)(2k+3)}{3}. \end{aligned}$$

Note: the step marked * is difficult to imagine doing, unless you have paid careful attention to the desired result. It is prompted by looking for the $(k + 1)$ term in the right-hand side of $S(k + 1)$.

Hence, by the principle of mathematical induction,

$$1^2 + 3^2 + 5^2 + \dots + (2n - 1)^2 = \frac{n(2n - 1)(2n + 1)}{3},$$

for all integers $n \geq 1$.

(b) *Proof:* Base Step: For $n = 1$, we have $1(2^1) = 2$ and $2 + (1 - 1)2^{1+1} = 2$, so the base case holds.

Induction step: For some integer $k \geq 1$, assume that the statement is true for $n = k$. That is, assume that

$$\sum_{i=1}^k i(2^i) = 2 + (k - 1)2^{k+1}.$$

(This is the induction hypothesis.)

Now we want to prove that the statement holds for $n = k + 1$. That is we want to prove that

$$\begin{aligned} \sum_{i=1}^{k+1} i(2^i) &= 2 + ((k + 1) - 1)2^{(k+1)+1} \\ &= 2 + k 2^{k+2}. \end{aligned}$$

We have

$$\begin{aligned} \sum_{i=1}^{k+1} i(2^i) &= \sum_{i=1}^k i(2^i) + (k + 1)2^{k+1} \\ &= 2 + (k - 1)2^{k+1} + (k + 1)2^{k+1} && \text{(By the induction hypothesis.)} \\ &= 2 + k 2^{k+1} - 2^{k+1} + k 2^{k+1} + 2^{k+1} \\ &= 2 + 2k 2^{k+1} \\ &= 2 + k 2^{k+2}. \end{aligned}$$

Hence, by the principle of mathematical induction,

$$\sum_{i=1}^n i(2^i) = 2 + (n-1)2^{n+1}$$

for all integers $n \geq 1$.

(c) *Proof:* Base Case: For $n = 1$, we have $2 = 1^2 + 1$, so this is true.

Induction Case: We wish to show that $S(k) \Rightarrow S(k+1)$ for some integer $k \geq 1$, where

$$S(k) \text{ is } 2 + 4 + \dots + 2k = k^2 + k$$

$$\text{and } S(k+1) \text{ is } 2 + 4 + \dots + 2k + 2(k+1) = (k+1)^2 + (k+1).$$

So consider the left-hand side of $S(k+1)$:

$$\begin{aligned} (2 + 4 + \dots + 2k) + 2(k+1) &= (k^2 + k) + 2(k+1) \\ &= k^2 + 3k + 2 \\ &= (k^2 + 2k + 1) + (k+1) \\ &= (k+1)^2 + (k+1). \end{aligned}$$

Hence, by the principle of mathematical induction,

$$2 + 4 + 6 + \dots + 2n = n^2 + n$$

for all integers $n \geq 1$.

$\Rightarrow$ 4. [M] Prove using mathematical induction that, for every integer $n \geq 2$,

$$\sum_{i=2}^n i(i-1) = \frac{n(n-1)(n+1)}{3}.$$

Solution:

Proof: Base Case: For $n = 2$, we have $2 \times 1 = 2$ and $\frac{2(2-1)(2+1)}{3} = 2$, so the statement holds for $n = 2$.

Induction Case: We wish to show that if the statement holds for $n = k$, for some integer $k \geq 2$, then it holds for $n = k + 1$. So assume that

$$\sum_{i=2}^k i(i-1) = \frac{k(k-1)(k+1)}{3}. \quad (1)$$

We need to show that

$$\sum_{i=2}^{k+1} i(i-1) = \frac{(k+1)k(k+2)}{3}. \quad (2)$$

So consider the left-hand side of (2)

$$\begin{aligned} \sum_{i=2}^{k+1} i(i-1) &= \sum_{i=2}^k i(i-1) + (k+1)k \\ &= \frac{k(k-1)(k+1)}{3} + (k+1)k \quad (\text{By the induction hypothesis (1)}) \\ &= \frac{k(k-1)(k+1)}{3} + \frac{3k(k+1)}{3} \\ &= ((k-1) + 3) \frac{k(k+1)}{3} \\ &= \frac{k(k+1)(k+2)}{3} \end{aligned}$$

Which is the right-hand side of (2). Hence, by the principle of mathematical induction,

$$\sum_{i=2}^n i(i-1) = \frac{n(n-1)(n+1)}{3}.$$

5. [M] A sequence of numbers $a_1, a_2, a_3, \dots$ is defined by

$$a_1 = 1, \quad a_2 = 2, \quad a_n = a_{n-1} + a_{n-2}$$

where n is an integer and $n \geq 3$.

- (a) Determine the values of a_3, a_4, a_5, a_6 and a_7 .

- (b) Using the principle of mathematical induction, prove that for all $n \geq 1$,

$$a_n < \left(\frac{7}{4}\right)^n.$$

Solution:

- (a) The values are: $a_3 = 3, a_4 = 5, a_5 = 8, a_6 = 13$, and $a_7 = 21$.

- (b) *Proof:* Base Case: The result holds for $n = 1$ and $n = 2$ as $1 < (7/4)^1$ and $2 < (7/4)^2 = 49/16$.

Induction Case: Assume that $a_n < (7/4)^n$ for all $n \leq k$, where $k \geq 2$ (strong induction hypothesis). We want to prove that

$$a_{k+1} < \left(\frac{7}{4}\right)^{k+1}.$$

From the definition of a_n , we have $a_{k+1} = a_k + a_{k-1}$.

By the strong induction hypothesis, $a_{k-1} < (7/4)^{k-1}$ and $a_k < (7/4)^k$. Hence

$$\begin{aligned} a_{k+1} &= a_k + a_{k-1} \\ &< \left(\frac{7}{4}\right)^k + \left(\frac{7}{4}\right)^{k-1} \\ &= \left(\frac{7}{4}\right)^{k-1} \left(\frac{7}{4} + 1\right) \\ &= \left(\frac{7}{4}\right)^{k-1} \left(\frac{11}{4}\right) \\ &= \left(\frac{7}{4}\right)^{k-1} \left(\frac{44}{16}\right) \\ &< \left(\frac{7}{4}\right)^{k-1} \left(\frac{49}{16}\right) \\ &= \left(\frac{7}{4}\right)^{k-1} \left(\frac{7}{4}\right)^2 \\ &= \left(\frac{7}{4}\right)^{k+1} \end{aligned}$$

Hence by strong induction, for all $n \geq 1$,

$$a_n < \left(\frac{7}{4}\right)^n.$$

$\Rightarrow$ 6. The following statements are **not** true. Try to prove that they are true using induction and explain where the induction proof breaks down in each case.

(a) [M] $1^2 + 2^2 + \dots + n^2 = \frac{2n^3 + 3n^2 + n + 6}{6}$ for all positive integers.

(b) [M] $5n = 0$ for all $n \in \mathbb{N}$.

Solution:

(a) This exercise illustrates what can happen if you do not check the base case.

Let $S(n)$ be the statement

$$1^2 + 2^2 + \dots + n^2 = \frac{2n^3 + 3n^2 + n + 6}{6}$$

for all positive integers.

Base Step: Let's suppose for now that we forget this step.

Induction Step: The induction step works in the usual way. Assume that $S(k)$ is true, that is

$$1^2 + 2^2 + \dots + k^2 = \frac{2k^3 + 3k^2 + k + 6}{6}$$

for some natural number k .

Now we prove that $S(k+1)$ is true. That is

$$1^2 + 2^2 + \dots + (k+1)^2 = \frac{2(k+1)^3 + 3(k+1)^2 + (k+1) + 6}{6}.$$

Considering the left-hand side of the equation above, we have

$$\begin{aligned}
 & 1^2 + 2^2 + \dots + k^2 + (k+1)^2 \\
 &= \frac{2k^3 + 3k^2 + k + 6}{6} + (k+1)^2 \quad \text{by the induction hypothesis} \\
 &= \frac{1}{6} (2k^3 + 3k^2 + k + 6 + 6(k+1)^2) \\
 &= \frac{1}{6} (2k^3 + 9k^2 + 13k + 12) \\
 &= \frac{1}{6} (2(k+1)^3 + 3(k+1)^2 + (k+1) + 6)
 \end{aligned}$$

which is $S(k+1)$.

The induction step works fine, however the statement is not true when $n = 1$, or for any n in fact. Therefore nothing we prove about the $(k+1)$ case makes any sense, regardless of whether it appears to work or not. The base case has to work otherwise the inductive step, which takes us from one case to the next, has no starting point.

- (b) This exercise illustrates what happens if you don't check for which values your induction step works/don't check the required base cases for strong induction.

Consider the statement

$$5n = 0$$

for all $n \in \mathbb{N}$.

Base Step: The statement holds for $n = 0$ since $5n = 5 \cdot 0 = 0$.

Induction Step: Assume that the statement holds for all $m \leq k$ for some $k \geq 1$, that is

$$5m = 0$$

for all $m \leq k$.

Now we want to prove that the statement holds for $n = k+1$. That is, that

$$5(k+1) = 0$$

We see that $5(k+1) = 5k + 5.1$. Furthermore, by the induction hypothesis, $5k = 0$, and $5.1 = 0$, and hence $5(k+1) = 5k + 5.1 = 0 + 0 = 0$.

The induction step again seems to work fine, however, it is clear that $5.1 \neq 0$. Our base case covered $5.0 = 0$, so our induction step should take us from the case $n = 0$ to the case $n = 1$. However, if we use the above argument, we see that in the inductive step, we would write 5.1 as $5.0 + 5.1$ and we haven't shown anything about 5.1 .

(Note that if we would have shown that the statement holds for both $n = 0$ and $n = 1$ our proof would be valid.)

7. [M] Recall that a real number x is *rational* if and only if it can be expressed as $x = \frac{a}{b}$ where a and b are integers and $b \neq 0$. A real number that is not rational is called *irrational*.

Let R be the statement: *The square root of any irrational number is irrational.*

- (a) Write a negation for R .
- (b) Prove R by contradiction.

Solution:

- (a) $\neg R$ is *There exists an irrational number whose square root is rational.*
- (b) *Proof:* Suppose there is such a number, x , such that $\sqrt{x} = \frac{a}{b}$ where a and b are integers and $b \neq 0$. Then

$$\begin{aligned} x &= \left(\frac{a}{b}\right)^2 \\ &= \frac{a^2}{b^2} \end{aligned}$$

But a^2 and b^2 are integers, and $b^2 \neq 0$, so x is rational, which contradicts our supposition.

Hence the square root of any irrational number is irrational.

8. [C] Prove that the difference of any rational number and any irrational number is itself irrational. Use proof by contradiction.

Solution:

Proof: To begin, we assume that there exists some x and y such that x and $x - y$ are rational, but y is irrational.

That is,

$$\begin{aligned}x &= \frac{a}{b} \\x - y &= \frac{c}{d},\end{aligned}$$

where a , b , c , and d are integers and b and d are not 0.

But then $y = x - (x - y)$, so

$$\begin{aligned}y &= \frac{a}{b} - \frac{c}{d} \\&= \frac{ad - bc}{bd}.\end{aligned}$$

But since $ad - bc$ and bd are integers, and $bd \neq 0$, y is rational. This contradicts our assumption.

Hence, the difference of any rational number and any irrational number must be itself irrational.